

Updated Architecture

Sovereign SLM for Aircraft Ground-Recovery Planning

Architecture principle

Aircraft systems detect and alert. Flight crew manages flight safety. The aircraft sends relevant event data to the ground. The Sovereign SLM accelerates ground-recovery planning. Authorized airline personnel make the final decision.

Scope

This architecture is for an AI-assisted maintenance and operational recovery workflow. The SLM does not determine aircraft airworthiness, decide whether a flight should continue or divert, or replace certified aircraft warning systems, flight-crew procedures, Maintenance Control, or authorized engineering decisions.

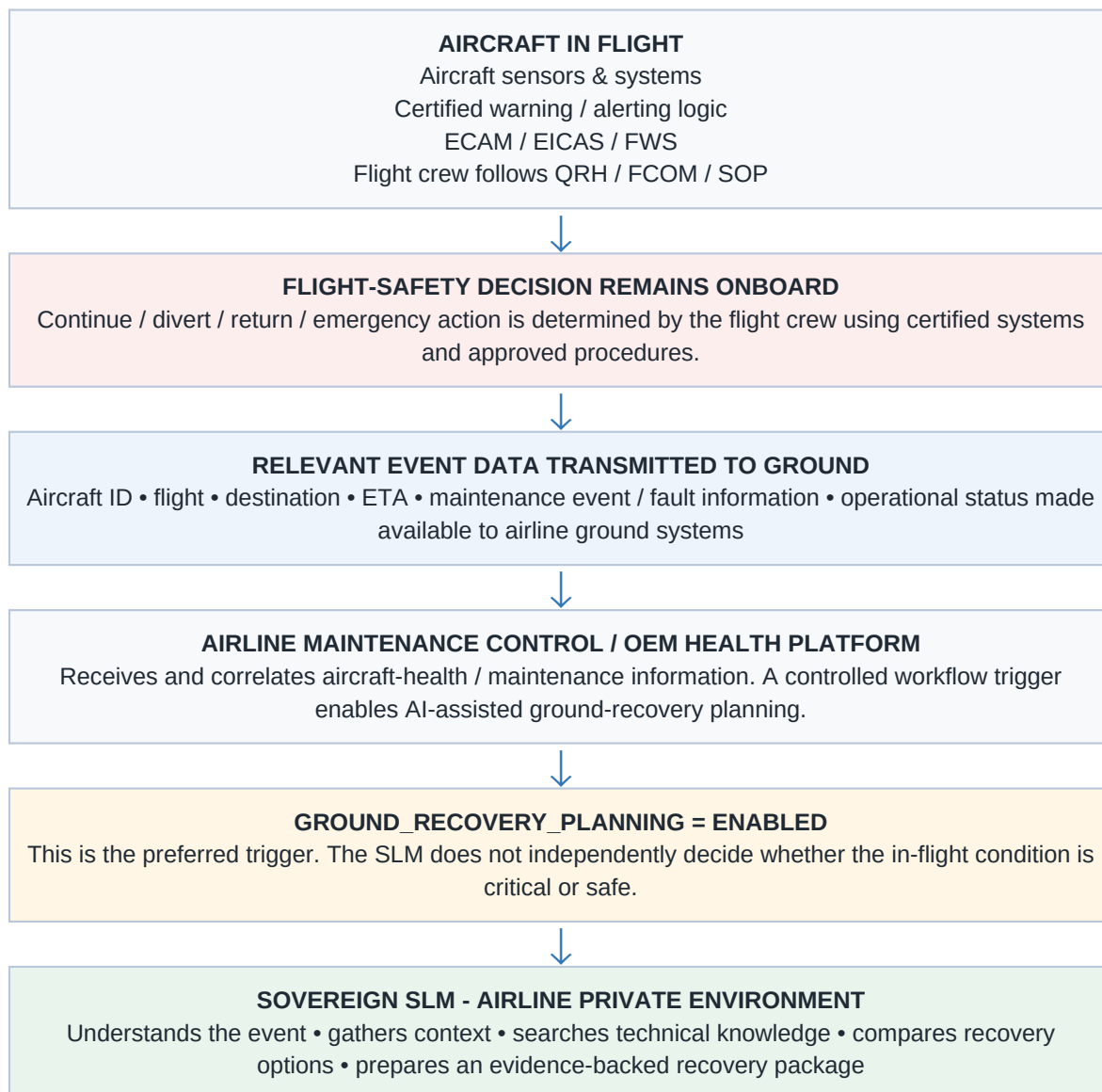
IN AIRCRAFT Detect & alert	GROUND AI Understand & orchestrate	HUMAN Authorize
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Illustrative airline scenario

An A350 operating a Dubai (DXB) to Frankfurt (FRA) sector transmits a maintenance event while airborne. The flight crew continues to manage the flight under approved procedures. The ground-based Sovereign SLM uses the remaining flight time to assemble maintenance, parts, people, and schedule information so the ground team can prepare before arrival.

1. Updated End-to-End Architecture

The SLM is intentionally located in the airline's ground data center, private cloud, or VPC. No SLM is required on the aircraft for this use case.



2. Ground-Based Sovereign SLM Platform

The model sits close to the airline systems it needs to use. This reduces integration complexity and keeps sensitive maintenance and operational data within airline-controlled infrastructure.



Design rule: the SLM reasons over airline information, but deterministic tools perform calculations that are better handled by rules or optimization engines.

3. Safety and Decision Boundaries

The architecture separates aviation safety decisions from AI-assisted recovery planning.

Zone	Examples	SLM role
GREEN	Search maintenance history; retrieve manuals; check parts; find technicians; compare ground-recovery scenarios; draft work order.	May execute read-only actions and draft recommendations through approved tools.
AMBER	Reserve a component; notify station team; propose aircraft swap; prepare an MEL-related post-landing option.	Human approval required before action.
RED	Airworthiness; diversion decision; emergency procedure; whether the aircraft should continue flight; final maintenance release.	Outside SLM authority. Existing certified systems, flight crew, MCC, and authorized engineering procedures apply.

Important trigger

The SLM should react to a controlled ground-side status such as **GROUND_RECOVERY_PLANNING = ENABLED**, rather than trying to classify the event itself as safe, unsafe, critical, or non-critical.

If the aircraft is involved in a serious event or diverts, the SLM may still help with ground logistics - such as locating maintenance capability, parts, stands, or recovery resources - but it must not influence the airborne safety decision.

4. Runtime Flow for the DXB-FRA Demo

1	Aircraft event occurs Existing certified aircraft systems detect the event. The flight crew receives the appropriate alert and follows approved procedures.
2	Flight crew manages the flight The decision to continue to Frankfurt, divert, return, or take emergency action remains with the flight crew and airline procedures.
3	Event reaches ground systems Relevant aircraft-health / maintenance data is transmitted to the airline ground environment while the aircraft is still airborne.
4	Ground recovery workflow is enabled Maintenance Control or an approved airline workflow authorizes AI-assisted ground-recovery planning.
5	Sovereign SLM gathers context The SLM queries maintenance history, technical documentation, parts inventory, engineer roster, flight rotation, and alternative aircraft availability.
6	Rules and optimization calculate impacts Deterministic services estimate part-transfer time, schedule impact, aircraft-swap consequences, and other operational calculations.
7	Recovery options are prepared The SLM presents evidence-backed options such as inspect/repair at FRA, prepare an MEL-related post-landing path for authorized review, move a part from another station, or evaluate an aircraft swap.
8	Human approves preparation Maintenance Control / authorized engineering personnel approve, modify, or reject the preparation plan. AI does not release the aircraft.
9	Aircraft lands Technician, tools, documents, parts contingency, and OCC awareness are already prepared so physical troubleshooting can start immediately.

5. Recommended Technology Pattern

For a prototype, keep the architecture simple and explicit. The goal is to demonstrate orchestration across multiple airline information sources - not to simulate certified aircraft systems.

Layer	Prototype choice	Production direction
Aircraft event feed	Synthetic JSON / webhook	Existing airline / OEM aircraft-health platform
SLM inference	Local 7B-14B model via vLLM or equivalent	Airline-controlled inference platform in private DC / VPC
RAG	Hybrid search over demo MEL / manuals / work orders	Versioned, access-controlled technical knowledge platform
Tool integration	MCP / controlled REST tools	Enterprise API / MCP gateway with RBAC and policy enforcement
Structured systems	PostgreSQL / mock APIs	MRO, ERP, OCC, roster, fleet, inventory systems
Optimization	Python rules / scheduling service	Dedicated deterministic optimization / OR services
Human workflow	Web approval screen	Integrated MCC / engineering workflow
Governance	Prompt / tool guardrails + audit log	RBAC, DLP, tool permissions, full traceability, observability

Architecture summary

Aircraft systems detect. The flight crew manages flight safety. The aircraft sends the relevant event to the ground. The Sovereign SLM understands and orchestrates. Deterministic systems calculate. Authorized humans decide.

For this Soar.AI use case, a ground-based Sovereign SLM is the preferred design because most required data - maintenance history, technical documentation, parts inventory, staff availability, flight rotations, and fleet options - already resides in airline ground systems.